Dr. Yonatan Ashenafi

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Applied mathematician specializing in stochastic modeling, statistical inference, and data-driven analysis of complex dynamical systems. Research spans biological transport, collective motion, and machine learning applications in scientific computing. Teaching experience covers calculus, differential equations, probability, and data mathematics at research universities.

Experience

Postdoctoral Fellow, Worcester Polytechnic Institute (WPI) Aug 2023 –
Worcester, MA

- Developing stochastic and simulation-based inference models for intracellular cargo transport using experimental data from multi-university collaborations (University of Vermont).
- Teaching Multivariable Calculus, Calculus I & II, and Ordinary Differential Equations.

Postdoctoral Fellow, University of Alberta Sep 2021 – May 2023
Edmonton, AB

- Constructed stochastic dynamical models to quantify collective motility mechanisms in microscopic algae from experimental video data.
- Taught Calculus I for Life Sciences and Calculus II for Physical Sciences.

Technical Fellow Intern, General Electric (GE) Research Oct – Dec 2020
GE Global Research Center, Niskayuna, NY

- Collaborated with the Probabilistic Design Team on Bayesian experimental design and additive manufacturing; resulting paper received the ASME CIE Best Paper Award (2022).
- Developed Python programs for reinforcement learning tasks.

Teaching Assistant, Department of Mathematical Sciences 2016 – 2021
Rensselaer Polytechnic Institute (RPI), Troy, NY

- Led recitation sections, office hours, and exam review sessions for undergraduate courses.
- Courses: Calculus I–III, Introduction to Differential Equations, Probability, Introduction to Data Mathematics, and Foundations of Applied Mathematics.

Research Assistant, Dordt College Jun – Aug 2014
Sioux Center, IA

- Constructed Gaussian mixture models from bacterial gene expression data using R.

Tutor, Dordt College Library Sep 2013 – May 2016
Sioux Center, IA

- Tutored students in College Algebra, Calculus I–III, Linear Algebra, and Discrete Mathematics.

Education

Rensselaer Polytechnic Institute (RPI), Troy, NY
Ph.D. in Applied Mathematics, May 2021
Thesis: *Stochastic Hydrodynamics of Colonial Microswimmers*

Dordt University, Sioux Center, IA
B.A. in Mathematics, May 2016

Publications

Stochastic Modeling and Mathematical Biology

- Ashenafi, Y. & Kramer, P. R. (2026). Stochastic Analysis of Kinesis and Taxis for Colonial Protozoa. *Bulletin of Mathematical Biology.*
- Ashenafi, Y. & Kramer, P. R. (2024). Statistical Mobility of Multicellular Colonies of Flagellated Swimming Cells. *Bulletin of Mathematical Biology*, 86, 125.
- Ashenafi, Y. (2022). Stability and Spatial Autocorrelations of Suspensions of Microswimmers with Heterogeneous Spin. *Physics of Fluids*, 34(12), 121908.
- Zheng, P., Kumadaki, K., Quek, C., Lim, Z. H., Ashenafi, Y., Yip, Z. T., Newby, J., Alverson, A. J., Jie, Y., & Jedd, G. (2023). Cooperative Motility, Force Generation and Mechanosensing in a Foraging Non-Photosynthetic Diatom. *Open Biology*, 13(10), 230148.

Machine Learning and Statistical Inference

- Ashenafi, Y. (Under review). Variational Dimension Lifting for Robust Tracking of Nonlinear Stochastic Dynamics. *Physica D.*
- Ashenafi, Y., Pandita, P., & Ghosh, S. (2022). Reinforcement Learning-Based Sequential Batch-Sampling for Bayesian Optimal Experimental Design. *Journal of Mechanical Design*, 144(9), 091705.
- Disselkoen, C., Greco, B., Cook, K., Koch, K., Lerebours, R., Viss, C., Cape, J., Held, E., Ashenafi, Y., Fischer, K., Acosta, A., Cunningham, M., Best, A. A., DeJongh, M., & Tintle, N. (2016). A Bayesian Framework for the Classification of Microbial Gene Activity States. *Frontiers in Microbiology*, 7, 1191.

In Preparation

- Ashenafi, Y. & Walcott, S. Simulation-Based Inference of Hidden Motor States in Intracellular Cargo Transport.

Awards

- Bill and Nancy Siegmann Applied Mathematical Modeling Prize, RPI (2021).
- ASME CIE Advanced Modeling and Simulation Best Paper Award (2022).
- DiPrima Summer Research Fellowship, RPI (2018).
- Viss Mathematics Scholarship (2016).

Teaching Interests

Stochastic Processes, Probability, Differential Equations, Mathematical Biology, Calculus (all levels), Linear Algebra, Data Mathematics, Machine Learning for Scientists.

Service and Activities

- Proposal Reviewer, National Science Foundation (NSF), 2025–present.
- Presenter, Canadian Mathematical Society Mathematics Education Meeting, 2025.
- Presenter, Biophysical Society Annual Meeting, 2025.
- Presenter, Biology and Medicine through Mathematics (BAMM) Conference, 2024.
- Participant, ICERM Workshop on Interacting Particle Systems: Analysis, Control, Learning, and Computation, 2024.
- Mathematical Modeler, Mathematical Problems in Industry (MPI) Workshop, Univ. of Vermont (2024) and RPI (2017).
- Judge, SIMIODE SCUDEM X Differential Equations Modeling Challenge, 2025.
- Organizer, Mini-Symposium on Noise and Asymmetry in Microscopic Life, CAIMS 2022.
- Presenter, Society for Industrial and Applied Mathematics (SIAM) Life Sciences Conference, 2021.
- Presenter, Mathematical Biosciences Institute Workshop, 2020.
- Presenter, University of Minnesota Multiscale Modeling Conference, 2019.
- Academic Consultant, Shaggar Institute of Technology, 2025–present.
- Presenter, Ethiopian Mathematics Professional Association Seminar, 2025.
- Judge, University of Alberta Undergraduate Research Symposium, 2022.
- Treasurer, Black Graduate Students Association, RPI (2020).

Technical Skills

Python (NumPy, Pandas, SciPy, TensorFlow, scikit-learn), MATLAB, C++, R, SQL

Professional Membership

Society for Industrial and Applied Mathematics (SIAM)

References

Dr. Peter Kramer
Professor, Rensselaer Polytechnic Institute
kramep@rpi.edu — 518-276-6896

Dr. Sam Walcott
Professor, Worcester Polytechnic Institute

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Dr. Michael Johnson
Teaching Professor, Worcester Polytechnic Institute
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